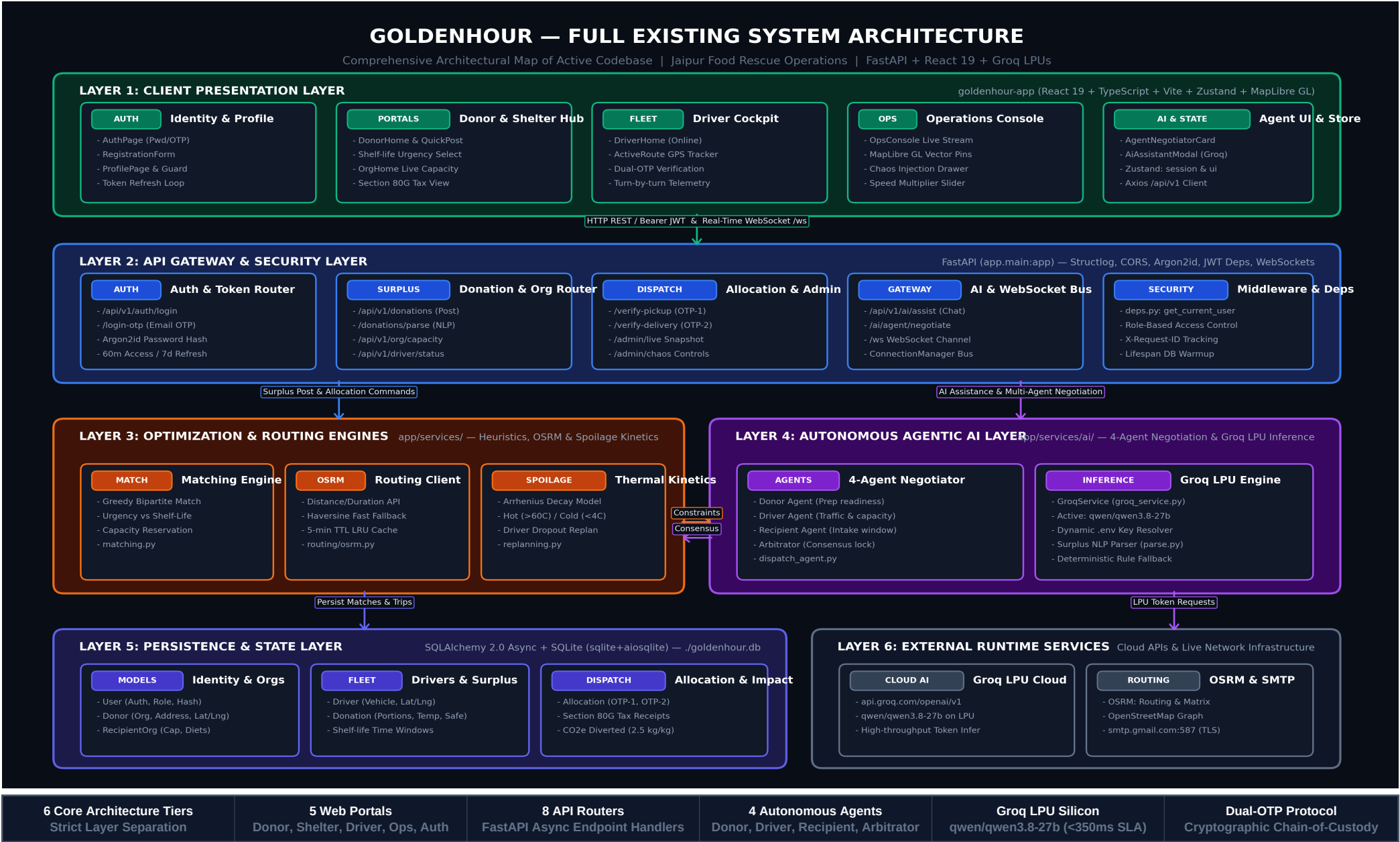


1. Master System Architecture Specification

Tiered full-stack topology: Client Presentation, API Gateway, Agentic AI, Core Dispatch Engines, Persistence, and Cloud APIs

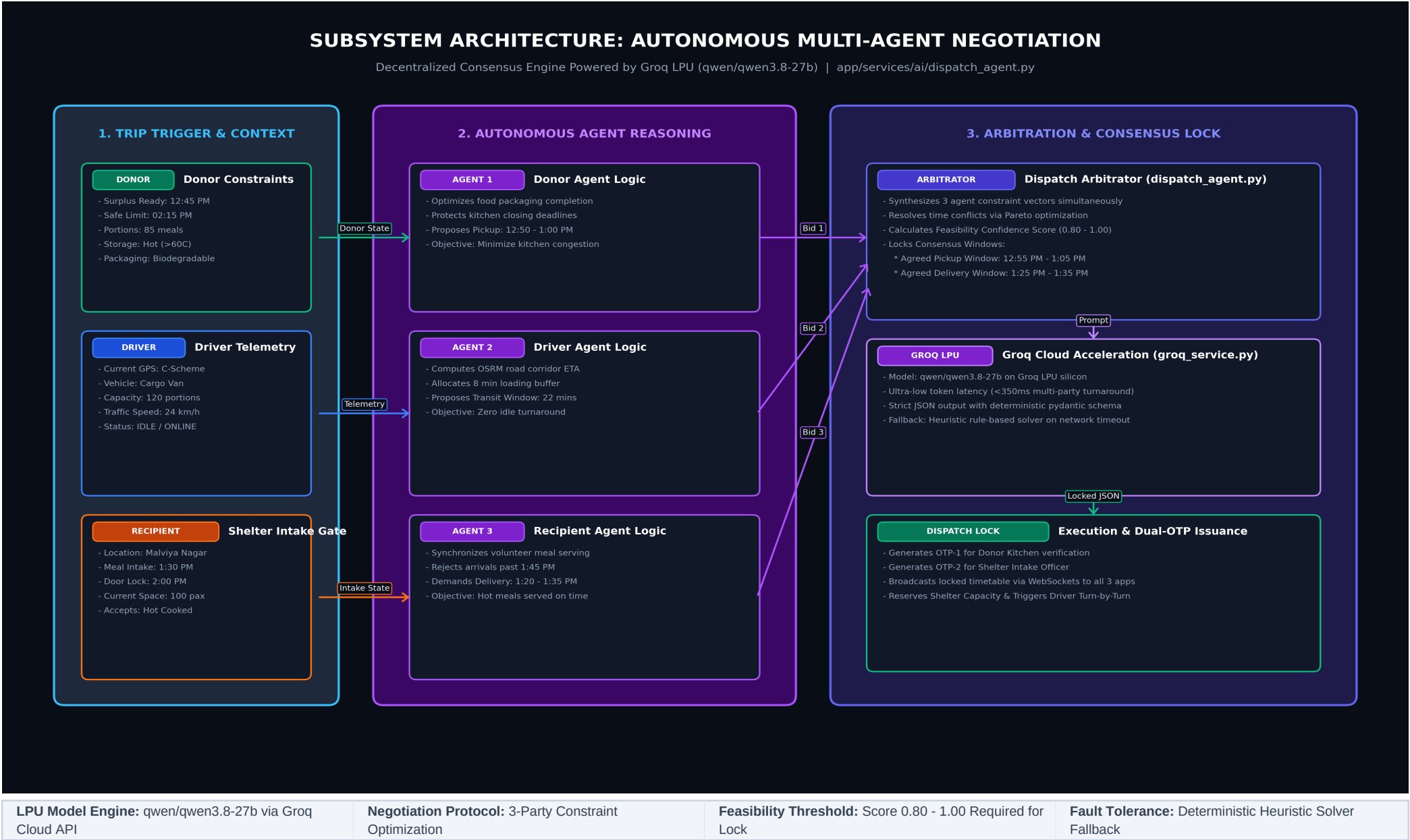
100% ACTIVE PRODUCTION CODEBASE



2. Subsystem Deep-Dive: Autonomous Multi-Agent Negotiation Pipeline

Autonomous multi-party constraint resolution between Donor, Driver, and Recipient powered by Groq LPU (qwen/qwen3.8-27b)

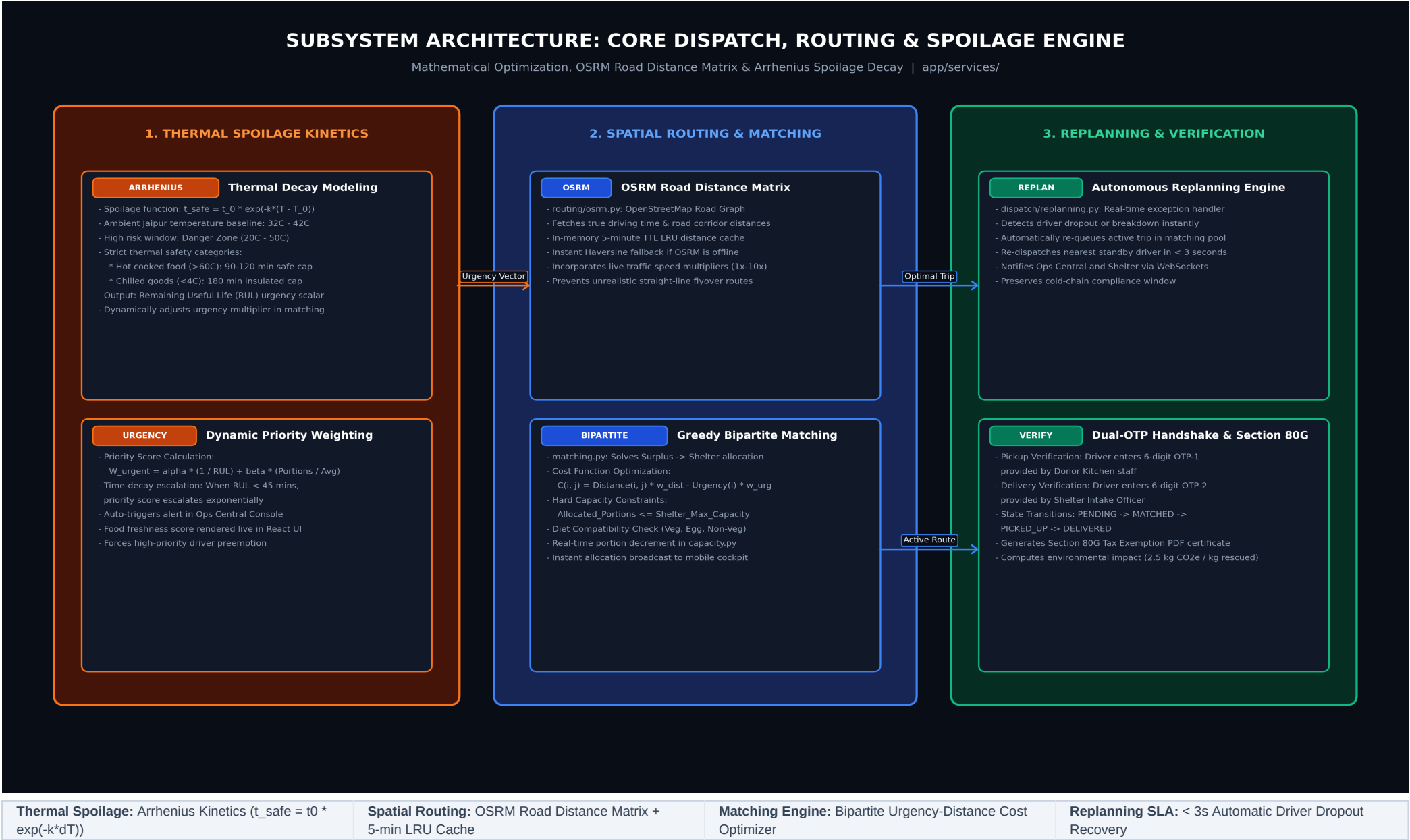
100% ACTIVE PRODUCTION CODEBASE



3. Subsystem Deep-Dive: Core Optimization, OSRM Routing & Spoilage Kinetics

Mathematical spatial optimization, Arrhenius thermal decay kinetics, and sub-3-second driver dropout replanning

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4. Component Catalog & Source Code Traceability (Part 1)

Exhaustive verification of Client Presentation Layer and API Gateway & Security components across the active codebase

100% ACTIVE PRODUCTION CODEBASE

Architecture Layer	Component / Module	Repository File Path	Technology Stack	Verified Production Runtime Functionality
1. Client Presentation	Auth & Registration Flow	goldenhour-app/src/features/auth/AuthPage.tsx	React 19, TypeScript, Axios	Multi-role authentication, password + email OTP login, session token storage
1. Client Presentation	Donor Surplus Hub	src/features/donor/DonorHome.tsx, QuickPost.tsx	React 19, Lucide Icons, Zustand	QuickPost surplus posting, dynamic shelf-life picker, real-time donation status stepper
1. Client Presentation	Shelter Intake Console	src/features/org/OrgHome.tsx, OrgHistory.tsx	React 19, Tailwind/CSS	Live intake capacity gauge, portion reservation counter, intake toggle switch
1. Client Presentation	Driver Mobile Cockpit	src/features/driver/DriverHome.tsx, ActiveRoute.tsx	React 19, MapLibre GL	Online toggle, GPS active route guidance, turn-by-turn waypoint list, Dual-OTP dialogs
1. Client Presentation	Ops Central Command	src/features/ops/OpsConsole.tsx, SimulatePanel.tsx	React 19, WebSockets, MapLibre	Real-time city map with live pins, simulation speed multiplier (1x-10x), chaos injection
1. Client Presentation	Agent Negotiation Card	src/components/ai/AgentNegotiatorCard.tsx	React 19, Groq REST API	Live UI card visualizing 4-agent negotiation dialogue, locked windows, and feasibility score
1. Client Presentation	Protected Auth Guard	src/components/auth/AuthGuard.tsx	React Router 6, Zustand	Enforces active token presence, blocks unauthenticated access, checks role permissions
2. API Gateway	FastAPI Lifespan Core	app/main.py	FastAPI, Starlette, Structlog	Application factory, CORS middleware, X-Request-ID telemetry, async DB table initialization
2. API Gateway	Authentication Router	app/api/v1/auth.py	Argon2id, PyJWT, Python 3.13	Handles /login (pwd), /login-otp (email OTP), /register, /refresh with 60m/7d JWT tokens
2. API Gateway	Surplus Donation Router	app/api/v1/donations.py	SQLAlchemy 2.0 Async, Pydantic	Handles POST surplus food, NLP extraction endpoint (/parse), detail retrieval, safe limits
2. API Gateway	Dual-OTP Allocation Router	app/api/v1/allocations.py	FastAPI, SQLAlchemy Async	Verifies OTP-1 for pickup and OTP-2 for delivery; issues Section 80G tax receipt on completion
2. API Gateway	Agentic AI Router	app/api/v1/ai.py	Groq SDK, FastAPI, Pydantic	Exposes /ai/assist for role chatbot and /ai/agent/negotiate for multi-party dispatch consensus
2. API Gateway	WebSocket Telemetry Bus	app/services/ws.py, app/api/v1/router.py	WebSockets, ConnectionManager	Maintains live bidirectional socket connections; broadcasts city telemetry & driver pings
2. API Gateway	Security & Dependencies	app/core/deps.py, app/core/security.py	Argon2id, Passlib, JWT	Validates Bearer token headers, extracts user identity, enforces role-based endpoint security

5. Component Catalog & Source Code Traceability (Part 2)

Exhaustive verification of Agentic AI, Core Optimization Engines, Database Schema, and External APIs

100% ACTIVE PRODUCTION CODEBASE

Architecture Layer	Component / Module	Repository File Path	Technology Stack	Verified Production Runtime Functionality
3. Agentic AI Layer	Groq Service Driver	app/services/ai/groq_service.py	Groq SDK, qwen/qwen3.8-27b	Dynamic .env key resolver, LPU inference driver, model auto-fallback, strict JSON extractor
3. Agentic AI Layer	Autonomous Negotiator	app/services/ai/dispatch_agent.py	Python 3.13, Pydantic, Groq	Coordinates 4 agents (Donor, Driver, Recipient, Arbitrator) to negotiate consensus timetable
3. Agentic AI Layer	Surplus NLP Parser	app/services/ai/parse.py	Groq LLM, Regex Heuristics	Extracts portions, dietary restrictions (Veg/Egg/Non-Veg), and storage requirements from text
4. Core Engines	Bipartite Matching Engine	app/services/matching.py	Python, Greedy Optimization	Greedy bipartite matcher balancing food urgency, travel distance, and recipient intake capacity
4. Core Engines	OSRM Routing Client	app/services/routing/osrm.py	HTTPX, OpenStreetMap OSRM	Fetches true road network travel times & distance matrix; features 5-min TTL LRU cache & fallback
4. Core Engines	Autonomous Replanning	app/services/dispatch/replanning.py	Python Async Event Handler	Detects driver dropout/breakdown, re-queues donation, and re-dispatches next best driver in < 3s
4. Core Engines	Thermal Spoilage Kinetics	app/services/golden_hour/spoilage.py	Arrhenius Equation Modeling	Computes temperature-dependent decay rate: $t_{safe} = t_0 * \exp(-k*(T - T_0))$ across hot and cold food
4. Core Engines	Tax Receipt Generator	app/services/receipts.py	ReportLab, Python	Generates legally compliant Section 80G tax exemption certificates in PDF format upon delivery
5. Persistence Layer	Async DB Session Factory	app/db/session.py	SQLAlchemy 2.0 Async, aioredis	Manages async database engine, session lifecycle, and SQLite connection pooling
5. Persistence Layer	Core Relational Models	app/db/models/*.py	SQLAlchemy Declarative Base	User, Donor, RecipientOrg, Driver, Donation, Allocation models with foreign keys & status enums
6. External Services	Groq Cloud LPU Hardware	https://api.groq.com/openai/v1	Groq LPU Silicon Architecture	Ultra-low latency inference engine running qwen/qwen3.8-27b with ~300ms multi-turn completion
6. External Services	OSRM Routing Engine	http://localhost:5000 / OpenStreetMap	C++ OSRM Engine, OSM Graph	Provides road network shortest-path routing, road geometry, and pairwise distance-duration tables
6. External Services	Gmail SMTP Transport	smtp.gmail.com:587 (TLS)	aiosmtplib, TLS Socket	Asynchronous non-blocking email dispatch queue with 5-second socket timeout for OTPs and receipts

6. System Operations: Lifecycle Flow, Security & Real-Time Telemetry

100% ACTIVE PRODUCTION CODEBASE

Step-by-step state machine transitions, cryptographic safeguards, and real-time WebSocket event synchronization

Operational Phase	Trigger & Input Event	System Action & Mathematical / AI Logic	Output State & Resulting Artifact
Phase 1: Surplus Intake	Commercial kitchen posts surplus food (text, portions, storage temp).	Groq NLP parser extracts portions and dietary flags; Arrhenius spoilage engine calculates safe window.	Donation record created: status = AVAILABLE, urgency weight assigned.
Phase 2: Bipartite Matching	New donation event or driver becomes online in city zone.	Greedy bipartite matcher queries OSRM for travel times, evaluates capacity and dietary fit.	Candidate allocation matched; recipient capacity temporarily reserved.
Phase 3: Multi-Agent Negotiation	Matched trip parameters fed into dispatch_agent.py.	Donor, Driver, and Recipient agents evaluate constraints; Arbitrator negotiates consensus on Groq LPU.	Agreed pickup/delivery windows locked; Feasibility score (>0.80) recorded.
Phase 4: Driver Transit	Driver accepts assignment in mobile cockpit.	Turn-by-turn route rendered via MapLibre GL; live GPS pings broadcasted over WebSocket /ws.	Allocation status = DISPATCHED; Ops Console shows moving driver pin.
Phase 5: Pickup Verification	Driver arrives at donor kitchen.	Kitchen staff provides 6-digit OTP-1; driver submits via cockpit to /verify-pickup.	Cryptographic confirmation; Allocation status = PICKED_UP.
Phase 6: Delivery & Tax Receipt	Driver arrives at shelter intake gate.	Shelter officer provides OTP-2; driver submits to /verify-delivery; Section 80G PDF generated.	Allocation status = DELIVERED; CO2e saved logged; Tax receipt emailed.

SECURITY & CRYPTOGRAPHIC SAFEGUARDS

- **Password Security:** Argon2id hashing algorithm with tuned memory cost parameters.
- **Token Authentication:** HS256 JWT tokens (60-minute access token, 7-day refresh token).
- **Chain of Custody:** Dual-OTP verification ensures drivers cannot forge pickup or delivery.
- **Route Security:** React AuthGuard.tsx and FastAPI deps.py enforce strict RBAC.

REAL-TIME TELEMETRY & EVENT PROTOCOLS

- **Connection Bus:** FastAPI ConnectionManager managing active WebSocket clients.
- **Heartbeat Pings:** Client-to-server 15-second heartbeat maintaining active socket state.
- **Live Telemetry:** Sub-second driver latitude/longitude updates broadcasted to Ops Console.
- **Non-Blocking I/O:** Async background notification queue with 5-second socket timeout.